My research concerns probabilistic and geometric techniques in stochastic growth models, a class of models in probability theory whose members are believed, but far from proven, to all exhibit certain universal features. This family of models is referred to as the KPZ universality class, for the physicists Kardar, Parisi, and Zhang who first described aspects of it [KPZ86].

A subclass of models thought to belong to this class is known as last passage percolation (LPP). In LPP models we have a *random* environment of rewards which paths, constrained to move in particular directions, collect. One object of interest is the maximum reward attainable over all paths with given endpoints, and it is an important goal to understand the random fluctuations of the maximum reward around its average.

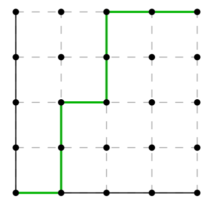

Fig. 1: *Paths in LPP.*

However, almost nothing is known about the fluctuations except in a handful of very particular types (or distributions) of randomness. For these choices of randomness, remarkable algebraic formulas facilitate the analysis, available due to connections to representation theory via special measures defined in terms of symmetric functions (eg. Schur [Oko01] or Macdonald measures [BC14]); this is the approach of *integrable probability*. But universality suggests that these conclusions should apply to a much wider class of distributions—where no integrable connections hold. It is of great interest to develop robust methods, and this is the direction my thesis work moves in. The approaches I pursue may broadly be classified as geometric and probabilistic.

Geometric aspects. The geometric thread of my work involves studying the *paths* in LPP that maximize the reward; such arguments will apply regardless of the type of randomness. Though these inferences currently rely on some crucial integrable inputs, this approach has already yielded results seemingly out of reach of integrable methods, such as the resolution of the long-standing "slow bond" problem [BSS14].

I highlight two projects in this line. In the first [BGHH20], I studied the geometry and maximum collective reward attained by *teams* of a fixed number of paths instead of a single one. Because of a faint resemblance to the fruit, the maximizing team is called a *geodesic watermelon*. Starting from natural assumptions about the LPP single path's maximum reward (and not about the randomness itself), I identified the width of the geodesic watermelon and the order of random fluctuations of the collective reward, developing new path properties of geodesic watermelons that hold deterministically for all choices of randomness. Surprisingly, a special case of these results translates back to the integrable setting and gives sharper results than what are directly available there (on the concentration of associated determinantal point processes).

The second project [GH20] continues the theme of geometric considerations yielding sharp results previously obtained via integrable methods. Under similar assumptions as in [BGHH20], I proved sharp tail probability exponents via purely geometric arguments; that is, I identify α such that the (upper tail) probability that the maximum reward is greater than x is approximately $\exp(-cx^{\alpha})$ for some constant c, and similarly for the lower tail probability that it is very small. This is a step towards universality: while the tail exponents are expected to be universal, there is currently no rigorous argument justifying it completely. [GH20] provides an important half of such an argument (the other half being to prove the assumptions hold for other distributions). The developed techniques may be also applicable in other non-integrable contexts like the notoriously difficult first passage percolation.

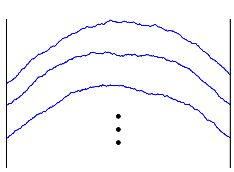

Fig. 2: *Parabolic Airy line ensemble*

Probabilistic aspects. The probabilistic thread of my dissertation concerns a central object in KPZ, the parabolic Airy line ensemble. This is a limiting object in many KPZ models and should be universal. It is an infinite collection of random non-intersecting functions; see Figure 2. My study of this ensemble has been via an important probabilistic resampling property it possesses, the *Brownian Gibbs property*, introduced in the fundamental paper [CH14]. This gives a valuable description of conditional probabilities in terms of Brownian motion, a ubiquitous probabilistic object that is extremely well-understood.

Partly because of this ubiquity, it was a natural belief that the parabolic Airy line ensemble should be strongly comparable with Brownian motion. A basic qualitative form of this comparison was proven in [CH14] using the Brownian Gibbs property. A major part of my dissertation work is to prove a strong *quantitative* comparison [CHH19] from limited integrable inputs; more precisely, I proved that the two processes' Radon-Nikodym derivative, a measure of distance of random processes, is very well-behaved, i.e., lies in all L^p spaces; this fact does not seem to be obtainable via integrable methods and is conjectured to be true more generally. With this fact one can estimate probabilities for the parabolic Airy line ensemble (quite difficult to do directly) in terms of Brownian motion (where they are within reach). These bounds give a deeper understanding of an important purportedly universal object and have already proven crucial in other studies in KPZ.

References

[BC14] Alexei Borodin and Ivan Corwin. Macdonald processes. *Probability Theory and Related Fields*, 158(1-2):225–400, 2014.

[BGHH20] Riddhipratim Basu, Shirshendu Ganguly, Alan Hammond, and Milind Hegde. Interlacing and scaling exponents for the geodesic watermelon in last passage percolation. *arXiv preprint arXiv:2006.11448*, 2020.

[BSS14] Riddhipratim Basu, Vladas Sidoravicius, and Allan Sly. Last passage percolation with a defect line and the solution of the slow bond problem. *arXiv preprint arXiv:1408.3464*, 2014.

[CH14] Ivan Corwin and Alan Hammond. Brownian Gibbs property for Airy line ensembles. *Inventiones Mathematicae*, 195(2):441–508, 2014.

[CHH19] Jacob Calvert, Alan Hammond, and Milind Hegde. Brownian structure in the KPZ fixed point. *arXiv preprint arXiv:1912.00992*, 2019.

[GH20] Shirshendu Ganguly and Milind Hegde. Optimal tail exponents in general last passage percolation via bootstrapping & geodesic geometry. *arXiv preprint arXiv:2007.03594*, 2020.

[KPZ86] Mehran Kardar, Giorgio Parisi, and Yi-Cheng Zhang. Dynamic scaling of growing interfaces. *Physical Review Letters*, 56(9):889, 1986.

[Oko01] Andrei Okounkov. Infinite wedge and random partitions. *Selecta Mathematica*, 7(1):57, 2001.